Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP-Based Chatbot

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Project Proposal Report

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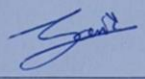
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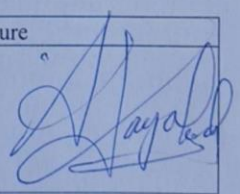
Declaration

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Digital learning platforms are widely used in modern education; however, many existing systems are not fully accessible to deaf and hard-of-hearing students. Most platforms rely on audio explanations and complex text-based materials, which can create difficulties for deaf learners, especially in technical subjects such as G.C.E. Ordinary Level (O/L) Information Technology.

To address this challenge, this research proposes an Emotion-Aware E-Learning System designed to support deaf O/L IT students. The system integrates an intelligent chatbot, real-time emotion detection, and adaptive learning content to provide interactive and personalized learning support. By analyzing students' emotional states and learning interactions, the system can deliver simplified explanations and visual learning materials that improve understanding and engagement.

The expected outcome is an accessible and intelligent learning platform that enhances learning experiences and supports inclusive education for deaf students.

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Keywords: *Emotion-Aware Learning, NLP Chatbot, Adaptive Learning, Deaf Education, Inclusive Education*

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
NLP	Natural Language Processing
CNN	Convolutional Neural Network
API	Application Programming Interface
REST	Representational State Transfer
FER	Facial Expression Recognition
O/L	Ordinary Level
IT	Information Technology
SUS	System Usability Scale
DSR	Design Science Research
SDG	Sustainable Development Goals
CoEAI	Centre of Excellence in Artificial Intelligence
HTTPS	Hypertext Transfer Protocol Secure
TLS	Transport Layer Security

1. INTRODUCTION

1.1. Background and Context

In recent years, digital learning platforms have significantly changed how students access education. Many schools and educational institutions now use online learning systems to deliver lessons, share learning materials, and support students outside the classroom. These platforms allow students to learn anytime and from any location. However, many existing e-learning systems are not fully accessible to all learners. Deaf and hard-of-hearing students still face many challenges when using most digital learning platforms.

Most e-learning systems rely heavily on audio explanations, spoken instructions, and large blocks of written text. These learning methods can be difficult for deaf learners because they mainly depend on visual information to understand lessons. As a result, deaf students may struggle to follow explanations, ask questions, or clearly understand complex learning materials.

This problem becomes more serious in subjects such as G.C.E. Ordinary Level (O/L) Information Technology (IT). IT is a technical subject that includes topics such as computer hardware, networking, operating systems, logic gates, and databases. These topics require clear explanations, visual demonstrations, and step-by-step guidance to help students understand complex technical concepts. However, many traditional e-learning systems do not provide enough interactive or visual learning support for deaf students. Without proper guidance, students may become confused or frustrated when learning difficult topics.

According to the World Health Organization (WHO), more than 430 million people worldwide live with disabling hearing loss, which represents over 5% of the global population. Many of these individuals are students who require accessible learning environments. In Sri Lanka, deaf students studying in special education schools or inclusive classrooms also face difficulties accessing digital learning materials that are mainly designed for hearing students. Therefore, improving accessibility in e-learning systems is an important step toward creating equal educational opportunities.

Several technologies have been introduced to support accessibility in digital learning, such as subtitles, captions, and visual learning materials. However, many existing learning systems still provide static content that does not change based on the student's understanding or learning progress. In addition, these systems do not consider the emotional state of the learner during the learning process. For example, if a student becomes confused or frustrated while studying a difficult IT concept, the system cannot detect the problem or provide additional support.

To address these limitations, this research proposes an intelligent emotion-aware e-learning system designed to support deaf O/L IT students. The system integrates several technologies, including facial emotion detection, adaptive learning mechanisms, secure communication infrastructure, and

an NLP-based educational chatbot. These components work together to create a more interactive and accessible learning environment.

The emotion detection component uses machine learning techniques to analyze students' facial expressions through a camera and identify emotional states such as engagement, confusion, or frustration. Based on these detected emotions, the adaptive learning component can automatically adjust the learning content. For example, the system may provide simpler explanations, additional visual materials, or alternative learning content to help students better understand the topic.

As part of this overall system, this research specifically focuses on developing an NLP-based intelligent chatbot subsystem. The chatbot uses Natural Language Processing (NLP) techniques to understand student questions written in natural language and provide clear and simplified explanations related to O/L IT topics. The chatbot also supports visual learning by providing diagrams, step-by-step explanations, and simple examples that are easier for deaf students to understand.

Furthermore, the chatbot integrates emotion-related information detected by the system. When the system identifies that a student is confused or frustrated, the chatbot can respond by providing simpler explanations, additional visual guidance, or alternative learning materials. In this way, the chatbot functions as an interactive learning assistant that helps students understand complex IT concepts and remain engaged during the learning process.

This research supports the United Nations Sustainable Development Goals (SDGs), particularly SDG 4: Quality Education and SDG 10: Reduced Inequalities. By improving accessibility and providing adaptive learning support, the system aims to reduce educational barriers for deaf students and create a more inclusive learning environment. Furthermore, this research belongs to the Artificial Intelligence research domain under the Centre of Excellence in Artificial Intelligence (CoEAI) research cluster. The project combines Natural Language Processing, emotion detection, adaptive learning techniques, and accessible learning design to support deaf O/L IT students more effectively.

1.2. Literature Review

Artificial Intelligence (AI) has increasingly been used to support inclusive education, particularly for students with hearing impairments. Researchers have explored how Natural Language Processing (NLP), intelligent tutoring systems, and visual learning technologies can improve accessibility for deaf learners. These technologies aim to reduce communication barriers and simplify complex learning content. However, most existing research treats these technologies as separate tools rather than as an integrated conversational assistant. This section reviews existing

research approaches, current products and services, and the limitations of existing systems related to NLP-based chatbot learning support.

Existing Research Approaches

NLP is widely applied to simplify technical educational content. Julianto et al. [1] developed an intelligent educational chatbot using transformer-based NLP models, demonstrating that such systems can deliver natural dialogue and accurate feedback to support independent learning. Similarly, Puspitasari et al. [2] highlighted that NLP-driven chatbots using text mining techniques help users find specific information more efficiently on digital platforms.

For learners with hearing impairments, multimodal Artificial Intelligence approaches are important. Fernando et al. [3] explored methods to support deaf learners in Sri Lanka by integrating emotion recognition and interactive communication technologies. Their study showed that combining different interaction methods, such as text and visual communication support, can help reduce communication barriers for deaf students in digital learning environments. Furthermore, Mohamed et al. [4] investigated Multi-Modal Fusion techniques that combine facial expression analysis and learner interaction patterns to personalize course content in virtual classrooms.

Emotion-aware learning systems have also emerged as an important research direction. Bala et al. [5] proposed a method for categorizing learners based on emotional states such as happiness or frustration using Convolutional Neural Networks (CNN). Additionally, Kaushik et al. [6] studied emotional support chatbots enhanced with deep learning techniques to interpret user emotions and provide personalized feedback during interactions.

Existing Products and Services

Several tools currently support deaf learners, although they still have limitations. Various assistive technologies have been developed to improve accessibility and help users better understand digital content through visual communication and simplified interaction methods. However, many of these systems mainly focus on communication assistance and do not provide interactive academic support for complex subjects.

E-learning platforms such as Moodle provide accessibility features including subtitles and captions for video-based learning materials [9]. Microsoft’s Reading Progress tool also provides visual feedback to help improve reading and literacy skills [10]. While these platforms support accessibility, they generally deliver static learning content and do not adapt based on a student's real-time emotional state, learning progress, or specific questions.

Limitations of Existing Systems

Despite these advancements, several gaps remain in existing systems:

- Lack of Emotion Awareness: Most chatbot systems cannot adjust their responses based on the emotional state of the learner [5], [6]. The proposed subsystem introduces Emotion-Aware NLP, where the chatbot's explanations and response style can change based on real-time facial emotion detection.
- Static Content Delivery: Many learning platforms provide subtitles or fixed explanations but do not dynamically adjust learning materials or provide additional visual aids such as diagrams when students experience difficulty understanding concepts [4].
- Curriculum Gaps: Many existing tools are designed for general education and are not specifically aligned with the Sri Lankan G.C.E. O/L IT syllabus. As a result, local students lack specialized academic support for this subject area [3].
- Limited Integration: Very few systems combine NLP-based conversational support, emotion detection, visual learning assistance, and adaptive responses within a single integrated learning subsystem [3], [4].

Summary

The literature shows that although AI and NLP technologies have improved accessibility in digital learning, most existing systems address only a single aspect of the problem. Few systems combine conversational learning support, emotion awareness, and adaptive visual explanations within one integrated platform. Therefore, this research proposes an NLP-based intelligent chatbot subsystem that provides simplified explanations, visual learning support, and adaptive responses based on real-time emotional feedback from students studying the G.C.E. O/L IT curriculum.

1.3. Research Gap

Our research addresses a significant gap in educational technology for deaf learners studying technical subjects, particularly within the Sri Lankan G.C.E. O/L IT curriculum. While various solutions exist such as NLP-based chatbots [1, 2], assistive communication tools [3], and emotion-

aware tutoring systems [4–6], there is currently no integrated system that combines conversational learning support with emotion-aware adaptive explanations for technical education.

Current systems are insufficient for several reasons:

- **Fragmented Functionality:** Most existing tools address only one aspect of accessibility. Some applications focus on communication assistance, while platforms like Moodle [9] mainly provide static captions and learning materials. These systems do not combine NLP-based content simplification, emotion detection, and visual learning support within a unified learning environment.
- **Lack of Conversational Emotion Adaptation:** Although some systems can detect learner emotions [5, 6], this information is typically used only for analysis. Current chatbot systems do not actively adapt their conversational responses based on a student's emotional state. For example, when a student shows frustration, existing systems rarely simplify explanations or switch to more visual learning methods.
- **Curriculum-Specific Gaps:** Many existing learning platforms are designed for general education and are not aligned with the Sri Lankan G.C.E. O/L IT syllabus. As a result, local deaf learners lack specialized digital learning support for understanding technical IT concepts.
- **Static Content Delivery:** Platforms providing captions or pre-recorded videos [9, 10] fail to dynamically trigger visual aids, such as diagrams or step-by-step tutorials, based on a learner's specific query or understanding level [4].

Justification of Novelty and Creativity

The novelty of this research lies in its Responsive Visual Tutoring Subsystem. Unlike traditional systems, this chatbot acts as a Multi-Modal Integration Hub. It creatively uses a feedback loop between text-based NLP and real-time facial expression analysis to provide a "Cognitive Intervention."

When the chatbot identifies "Confusion" during a technical IT lesson, it doesn't simply repeat the information; it re-routes the dialogue. It automatically simplifies the linguistic complexity and "pushes" interactive diagrams and step-by-step visual tutorials directly to the student. This ensures the chatbot functions as an Active Tutor that understands not just what the student is asking, but how they are feeling about the answer.

Table 1: Comparison of researches with the proposed solution

Reference	Integrated NLP + Visual + Emotion	Curriculum-Specific IT Support	Real-Time Emotion Adaptation	Dynamic Visual Aids
[1] Julianto et al.	No	No	No	No
[2] Puspitasari et al.	No	No	No	No
[3] Fernando et al.	Partial	No	Partial	No
[4] Mohamed et al.	Partial	No	Partial	Partial
[5] Bala et al.	No	No	Yes	No
[6] Kaushik et al.	No	No	Yes	No
Proposed Subsystem	Yes	Yes	Yes	Yes

In this table:

- "Yes" indicates the system addresses the specified challenge.
- "Partial" indicates the system addresses some aspects but not fully.
- "No" indicates the system does not address the challenge.

1.4.Research Problem

The proposed research introduces an NLP-based adaptive learning chatbot designed specifically for deaf students studying the Sri Lankan G.C.E. O/L IT syllabus. Unlike traditional educational chatbots that provide general responses, this system integrates emotion and difficulty-level detection to better understand the learning state of the student. By analyzing the student's text input, the chatbot can identify whether the student is confused or struggling and provide simplified explanations and additional guidance accordingly.

Another key contribution is the personalized learning support provided by the system. The chatbot adapts its responses based on the student's level of understanding, helping learners grasp complex IT concepts more easily. Furthermore, the system focuses on improving accessibility in digital education by offering an interactive learning assistant that supports deaf students in overcoming communication barriers. This approach aims to create a more inclusive and supportive learning environment.

1.5. Novel Contribution

The core focus of this research is the development of an NLP-driven Intelligent Tutoring Subsystem that functions as the central logic component of an emotion-aware e-learning platform. While traditional e-learning tools primarily provide static learning materials, deaf students studying the technical G.C.E. O/L IT syllabus require a system capable of simulating aspects of human tutor interaction. Therefore, this research focuses on designing a multi-modal integration framework that can interpret textual queries together with real-time emotional cues, such as confusion or frustration, in order to provide adaptive learning responses.

A key challenge lies in the semantic mapping of technical content. NLP models must be capable of simplifying complex IT concepts, such as networking layers or logic gate operations, into accessible explanations while maintaining academic accuracy. This requires the chatbot to move beyond simple keyword matching and instead apply intent recognition techniques to understand the underlying cause of a student's difficulty.

Another major challenge is maintaining real-time synchronization between emotion detection and conversational logic. Cognitive and emotional states in learners can change rapidly when they encounter unfamiliar technical terminology. If the system fails to effectively link a detected emotional state, such as confusion, with the chatbot's instructional response, learners may experience increased cognitive load. Consequently, the system must establish a continuous feedback mechanism in which the chatbot dynamically triggers visual learning aids such as diagrams, flowcharts, or interactive explanations based on real-time assessments of the student's comprehension.

Furthermore, there is currently limited availability of unified, curriculum-aligned digital tutoring systems designed specifically for deaf learners in Sri Lanka. Existing solutions often provide isolated functionalities, such as communication support tools or general-purpose NLP chatbots [1], [2], which lack the pedagogical adaptation required to support the specialized G.C.E. O/L IT curriculum.

By developing an integrated, responsive, and curriculum-focused chatbot subsystem, this research aims to address the challenge of providing accessible and adaptive learning support for deaf students. The objective is to move beyond passive information delivery and toward an interactive tutoring approach that enhances conceptual understanding, maintains learner engagement, and supports equitable educational outcomes through intelligent and visually oriented learning assistance.

2. OBJECTIVE

2.1 Main Objective

The primary objective of this research is to develop an intelligent emotion-aware chatbot subsystem that utilizes NLP and multi-modal data integration to provide adaptive and curriculum-aligned learning support for deaf students studying the G.C.E. O/L IT syllabus.

The system will integrate conversational AI, real-time emotion detection, and visual learning aids to generate personalized instructional responses that improve student comprehension and engagement.

2.2 Specific Objectives

1. **To develop a specialized NLP-based text simplification module** capable of interpreting G.C.E. O/L IT terminology and converting complex technical concepts into simplified and accessible explanations tailored for deaf learners.

2. **To receive real-time emotional data from the Real-Time Emotion Detection Subsystem** and categorize learner emotional states specifically confusion, frustration, and engagement during the learning process to enable adaptive responses.

3. **To retrieve and provide relevant visual learning content from the Adaptive Learning Content Subsystem,** including interactive diagrams, flowcharts, and step-by-step animations, based on student queries and detected emotional states.

4. **To integrate the NLP module, emotional data, and adaptive visual content into a unified chatbot architecture** that enables multi-modal interaction, real-time adaptive feedback, and personalized tutoring support.

5. **To evaluate the educational effectiveness and usability of the chatbot subsystem** by measuring improvements in learner comprehension, engagement, and satisfaction compared with traditional static e-learning methods.

3. METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research (DSR) approach, which is widely used in information systems and artificial intelligence research to design, develop, and evaluate technological solutions. DSR emphasizes the creation of innovative artifacts that solve real-world problems while generating new knowledge.

In this study, the proposed artifact is an Intelligent Emotion-Aware Chatbot Subsystem designed to support deaf students studying the G.C.E. O/L IT syllabus. The research process includes the following stages: problem identification, requirement analysis, system design, development, implementation, and evaluation.

The research begins by analyzing the challenges deaf learners face in understanding technical concepts through traditional static e-learning platforms. Based on these insights, the Chatbot Subsystem is designed to integrate NLP, real-time emotion detection, and visual learning support.

Crucially, the Chatbot Subsystem will receive real-time data from the Real-Time Emotion Detection Subsystem and content recommendations from the Adaptive Learning Content Subsystem. This enables the system to deliver personalized and adaptive learning experiences, responding dynamically to both the learner's emotional state and their knowledge needs.

Experimental testing and user-based evaluations will be conducted to measure the effectiveness of the system in improving learning comprehension, engagement, and accessibility for deaf students.

3.2 Technical Methods and Algorithms

The proposed chatbot subsystem integrates multiple AI techniques through a multi-modal data fusion architecture to provide adaptive learning support.

Natural Language Processing (NLP)

- The chatbot will use NLP to interpret student queries and generate simplified explanations for technical concepts in the G.C.E. O/L IT syllabus.
- Transformer-based models (e.g., BERT or GPT-based fine-tuned models) will perform intent recognition, semantic analysis, and text simplification.
- Techniques such as text mining, semantic mapping, and intent classification will ensure responses are accurate, clear, and accessible for deaf learners.

Real-Time Emotion Integration

- Emotional state data will be received from the Real-Time Emotion Detection Subsystem.
- The subsystem uses CNNs with Haar Cascade or MTCNN for face detection and ResNet or VGG for expression classification.
- Emotional states such as confusion, frustration, and engagement are transmitted as metadata to the Chatbot to adapt instructional responses.

Dynamic Visual Learning Support

- The Chatbot will trigger visual learning aids including diagrams, flowcharts, and step-by-step animations based on queries, emotional states, and recommendations from the Adaptive Learning Content Subsystem.
- This ensures that learners receive contextually relevant, multi-modal explanations for complex IT concepts.

Multi-Modal Integration and Decision Logic

- The Chatbot integrates NLP outputs, emotion metadata, and visual learning recommendations through a rule-based adaptive engine**.**
- Example: If a learner shows confusion while asking about a technical concept, the Chatbot retrieves additional visual aids from the Adaptive Learning Content Subsystem and delivers a simplified explanation, dynamically responding to both content and emotional context.

3.3 Tools and Platforms

Several tools and platforms will be used to develop and implement the proposed system.

Table 2: Tools & Platforms

Component	Tool / Platform	Justification
Programming Language	Python	Provides strong support for AI, NLP, machine learning, and data processing
NLP Framework	Hugging Face Transformers, spaCy, NLTK	Transformer-based models and advanced language processing
Deep Learning Framework	TensorFlow / PyTorch	Used to train and implement deep learning models
Computer Vision	OpenCV	Supports real-time facial image capture and processing
Backend Framework	Flask / FastAPI	Lightweight frameworks suitable for chatbot API development
Frontend Development	React.js and JavaScript	Used to build an interactive chatbot user interface
Database	MongoDB	Stores chatbot responses, learning resources, and interaction logs
Visual Content Design	Canva / Adobe Illustrator	Used to create educational diagrams and visual learning materials
Secure Communication / API Layer	RESTful API / WebSocket	Receive real-time data from Real-Time Emotion Detection and Adaptive Learning Content subsystems

These tools are selected because they provide reliable AI development environments, strong community support, and compatibility with modern machine learning frameworks.

3.4 Data Collection and Ethical Considerations

Data required for the chatbot subsystem will be collected from multiple sources to support system development and evaluation.

Educational content related to the G.C.E. O/L IT syllabus will be collected from textbooks, past examination papers, curriculum documents, and teacher notes. These materials will be used to train the NLP-based text simplification module and develop the chatbot knowledge base.

For emotion recognition, publicly available facial expression datasets such as FER-2013 will be used to train and validate the deep learning model. Additional testing data may be collected from volunteer students during the system evaluation phase.

Since the system involves facial expression analysis, ethical considerations are essential. All participants involved in testing will be informed about the purpose of the research, and informed consent will be obtained before collecting any data. For underage participants, consent will also be obtained from parents or guardians.

To protect user privacy, the system will not permanently store facial images. Instead, only the detected emotional category will be processed in real time. All collected data will remain confidential and will be used only for academic research purposes.

3.5 Validation Strategy and Performance Metrics

The effectiveness of the proposed chatbot subsystem will be evaluated using both technical performance metrics and user-based evaluation methods.

System Performance Evaluation

The emotion detection module will be evaluated using machine learning metrics such as accuracy, precision, recall, and F1-score to measure how effectively the system identifies emotional states. The emotion detection model is expected to achieve a minimum classification accuracy of 80%. For the NLP module, the quality of simplified explanations generated by the system will be evaluated using BLEU and ROUGE scores, which compare the chatbot's responses with expert-generated explanations. In addition, system response time will be measured to ensure that the chatbot provides real-time responses for student queries.

User Evaluation

User-based evaluation will be conducted with deaf students to measure the effectiveness of the chatbot in supporting learning. A pre-test and post-test approach will be used to compare student comprehension levels before and after interacting with the chatbot system. In addition, a System Usability Scale (SUS) questionnaire will be used to measure user engagement, satisfaction, and ease of use.

The results of these evaluations will help determine whether the proposed chatbot subsystem improves learning comprehension, engagement, and accessibility compared with traditional static e-learning methods.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed Emotion-Aware E-Learning System is designed to support deaf students studying the G.C.E. O/L IT syllabus through an intelligent and adaptive learning environment. The system integrates multiple AI technologies including NLP, facial emotion recognition, and adaptive learning algorithms to provide personalized learning support.

The architecture consists of three major intelligent subsystems: the Real-Time Emotion Detection Subsystem, the Adaptive Learning Content Subsystem, and the NLP Intelligent Chatbot Subsystem. These subsystems operate together through a centralized Application Server / Backend API Layer, which manages communication, data processing, and integration between the different components of the system.

The Emotion Detection Subsystem analyzes the student's facial expressions captured through the webcam to detect emotional states such as confusion, frustration, or engagement. Based on these emotional signals, the Adaptive Learning Content Subsystem dynamically modifies the learning materials by providing simplified explanations, visual diagrams, and step-by-step demonstrations. The NLP Intelligent Chatbot Subsystem acts as an interactive tutor that answers student questions by combining the student's query, emotional state, and learning context to generate appropriate responses.

Additionally, the system includes a secure data communication framework that ensures safe and efficient transmission of information between the client interface and the backend server using encrypted communication protocols.

4.2 High-Level System Architecture Diagram

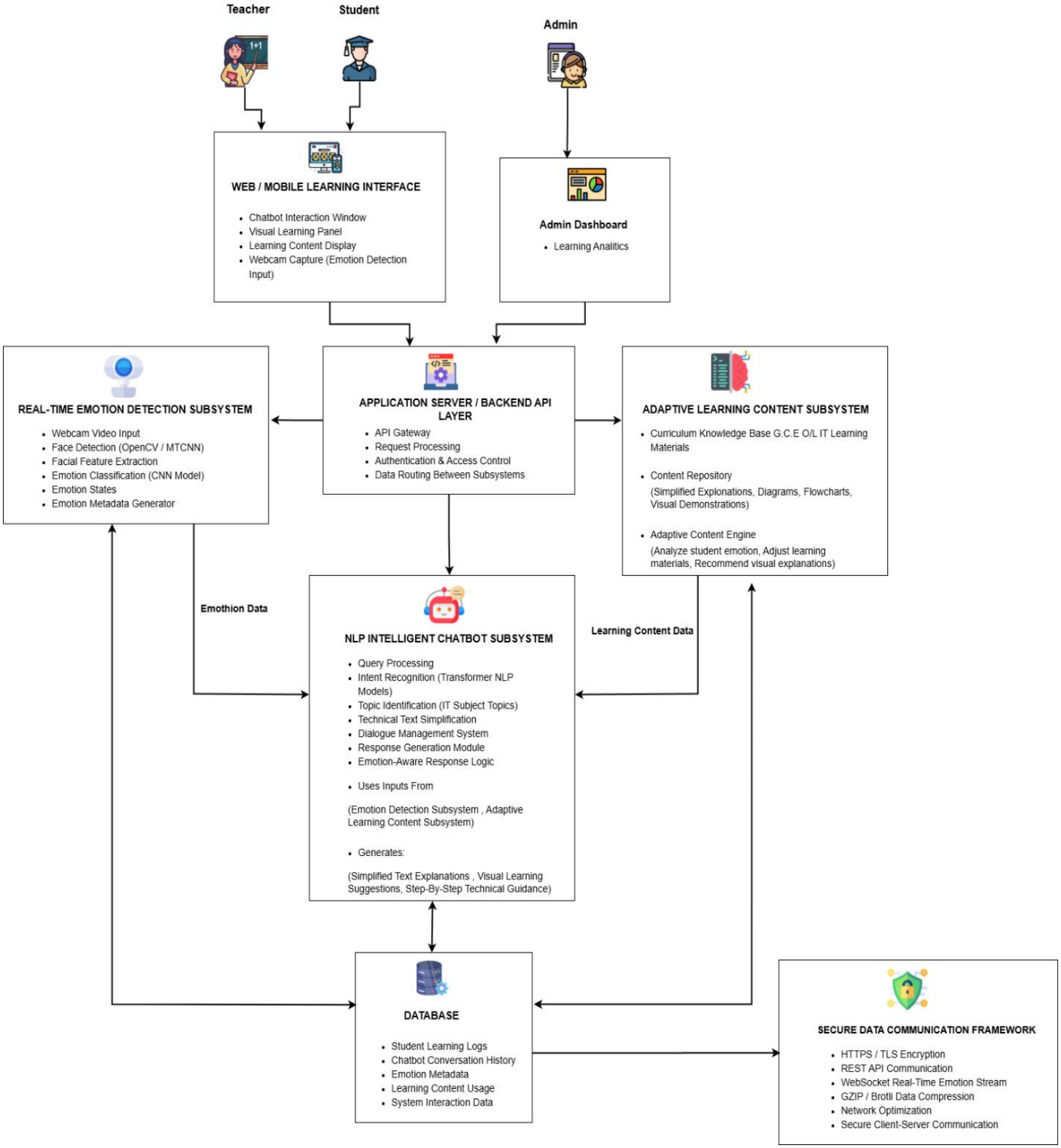


Figure. 1: System Architecture

4.3 Individual System Components and Their Functions

1. Web / Mobile Learning Interface

The Web/Mobile Learning Interface is the main platform where students and teachers interact with the system. It provides a chatbot window, visual learning panel, and learning content display. The interface also captures webcam input from the student device, which is used by the emotion detection subsystem.

2. Application Server / Backend API Layer

The Application Server manages communication between the user interface and system subsystems. It processes user requests, handles authentication, and routes data between the emotion detection, adaptive learning, and chatbot components.

3. Real-Time Emotion Detection Module

This subsystem analyzes facial expressions from the student's webcam using computer vision and deep learning models. It detects emotional states such as confusion, frustration, and engagement, and sends this emotion metadata to the system.

4. Adaptive Learning Content Module

The Adaptive Learning Content Subsystem adjusts learning materials based on student emotions and interactions. It provides simplified explanations, diagrams, flowcharts, and tutorials related to the G.C.E. O/L IT syllabus.

5. NLP Intelligent Chatbot Module

The chatbot processes student questions using NLP techniques such as intent recognition and text simplification. It uses emotion data and learning content information to generate context-aware explanations.

6. Database Layer

The database stores student interaction logs, chatbot conversations, emotion metadata, and learning content data. This information supports system analytics and learning improvement.

4.4 Component Interactions

The interaction between system components begins when a student accesses the learning platform through the Web/Mobile interface. The interface sends user queries and webcam data to the backend application server. The webcam stream is processed by the emotion detection subsystem, which identifies the student's emotional state.

The backend server then forwards the emotion metadata to the adaptive learning subsystem and the chatbot subsystem. The adaptive learning subsystem selects appropriate learning materials based on the student's emotional state and learning context. If the student asks a question, the chatbot processes the query using NLP techniques and generates an explanation using both emotion data and adaptive learning content.

The generated response is then returned to the user interface, where the student receives personalized explanations, diagrams, or step-by-step guidance.

4.5 Data Flow Explanation

The system data flow begins with the student interacting with the learning interface. User queries and webcam inputs are transmitted to the backend server. The backend routes webcam data to the emotion detection subsystem, where facial expression analysis is performed.

The resulting emotion metadata is transmitted back to the backend server and shared with the adaptive learning subsystem and chatbot subsystem. The adaptive learning subsystem selects suitable learning resources based on the detected emotional state. Meanwhile, the chatbot subsystem processes the student's question and combines query information, emotional data, and learning content to generate a personalized response.

The response is then delivered back to the learning interface, where it is displayed to the student in the form of simplified explanations, visual aids, or guided instructions.

4.6 Scalability and Security Considerations

To support a large number of users and ensure reliable system performance, the architecture is designed with scalability in mind. The backend API layer can be deployed on scalable cloud infrastructure, allowing the system to handle multiple simultaneous student sessions without significant performance degradation.

Security is a critical aspect of the system because it processes sensitive user interaction data and emotion-related information. All communication between the client interface and the backend server is secured using HTTPS and TLS encryption protocols. Additionally, optimized communication techniques such as REST APIs, WebSocket-based real-time streaming, and data compression methods including GZIP or Brotli are used to improve network efficiency and reduce bandwidth consumption.

These security and performance mechanisms ensure that the system can operate reliably in real-world educational environments while protecting user data and maintaining system responsiveness.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed Emotion-Aware Chatbot Subsystem involves several stakeholders who interact with or support the system in different ways.

Primary Users

- **Deaf and Hard-of-Hearing Students:**
  These students are the primary users of the system. They are studying the G.C.E. O/L IT syllabus and require simplified technical explanations and visual learning support to better understand complex concepts.

Secondary Stakeholders

- **Special Education Teachers / Instructors:**
  Teachers provide curriculum-aligned learning materials and verify the accuracy of simplified explanations generated by the chatbot. Their feedback helps ensure that the content is educationally appropriate and understandable for deaf students.

- **System Administrators:**
  System administrators are responsible for maintaining the chatbot backend infrastructure and managing the MongoDB database. They ensure that the system operates efficiently, securely, and without interruptions.

- **Developers (Project Team Members):**
  Developers are responsible for designing, developing, and maintaining the different subsystems that integrate with the chatbot, including the Emotion Detection Subsystem, Adaptive Learning Content Subsystem, and communication interfaces.

5.2 Functional Requirements

The functional requirements describe the main capabilities and services that the chatbot subsystem must provide.

FR1 – Textual Query Input: The system shall allow students to submit natural language questions related to the G.C.E. O/L IT syllabus through a web or mobile chat interface.

FR2 – Intent Recognition: The chatbot shall analyze user queries using NLP techniques to identify the specific IT topic (e.g., Logic Gates, Networking, Databases).

FR3 – Technical Text Simplification:
The system shall convert complex technical terminology into simplified explanations suitable for the literacy level of deaf students.

FR4 – Emotion Data Integration:
The chatbot shall receive and process real-time emotional metadata (such as confusion, frustration, or engagement) from the Real-Time Emotion Detection Subsystem.

FR5 – Dynamic Visual Triggering:
The system shall automatically retrieve and display visual learning aids, such as diagrams, flowcharts, or animations, from the Adaptive Learning Content Subsystem when confusion or difficulty is detected.

FR6 – Multi-modal Response Generation:
The chatbot shall combine simplified textual explanations with relevant visual learning resources to generate a clear and comprehensive response for the student.

FR7 – Interaction Logging:
The system shall record student–chatbot interactions and associated emotional metadata in the MongoDB database for learning analysis and system improvement.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the system.

NFR1 – Performance (Latency):
The system shall process user queries and generate responses within two seconds to ensure smooth real-time interaction.

NFR2 – Usability:
The chatbot interface shall be visually oriented, using clear layouts, high-contrast colors, and minimal complex text to support the learning needs of deaf students.

NFR3 – Accuracy:
The NLP module should achieve a minimum accuracy of 80% in intent recognition and provide contextually correct simplified explanations.

NFR4 – Security and Privacy:
The system shall not permanently store facial images of students. Only anonymized emotional metadata will be processed. All communication between system components shall be secured using encrypted protocols such as HTTPS.

NFR5 – Availability:
The chatbot subsystem shall be accessible 24 hours a day to support independent learning and continuous student access.

NFR6 – Scalability:
The backend architecture shall support multiple simultaneous users and maintain stable system performance without degradation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed Emotion-Aware E-Learning System includes educational institutions and organizations that support deaf and hard-of-hearing students. This includes special education schools, inclusive schools, and educational training centers that provide learning support for students with hearing impairments.

The system is specifically designed for students studying the G.C.E. O/L IT syllabus**,** where technical concepts often require clear explanations and visual learning materials.

A typical customer persona is a 15–16-year-old deaf student preparing for the G.C.E. O/L IT examination**.** These students prefer visual learning methods such as diagrams, simplified explanations, and step-by-step demonstrations instead of long text-based lessons.

Secondary users include teachers and school administrators who want to improve learning accessibility and monitor student learning progress through analytics.

6.2 Value Proposition

The proposed system provides an intelligent learning platform specifically designed to improve accessibility for deaf students. Traditional e-learning systems rely heavily on text or audio explanations, which can be difficult for deaf learners to understand.

This platform offers several key benefits:

- **Emotion-aware learning support** that detects student emotions during learning sessions.
- **Adaptive learning content** that adjusts explanations and visual materials based on the student's understanding.
- **Interactive AI chatbot tutor** that answers student questions using simplified explanations.
- **Visual learning materials** such as diagrams, flowcharts, and step-by-step demonstrations.

By combining emotion detection, adaptive learning, and NLP-based chatbot support, the system creates a more personalized and inclusive learning environment for deaf students.

6.3 Revenue Model

The system can generate revenue through several simple models suitable for an educational platform.

- **Subscription Model:**
  Educational institutions can subscribe to the platform by paying a yearly or monthly fee to allow their students to use the system.
- **Institutional Licensing:**
  Schools or educational centers can purchase a license that allows multiple students and teachers to access the system.
- **Freemium Model:**
  Basic learning support and chatbot features can be provided for free, while advanced features such as detailed learning analytics, additional visual resources, and advanced AI support can be included in a premium plan.

6.4 Cost Estimation

The development and operational costs of the system include several components as shown below.

Table 3: Cost Estimation

Category	Estimated Cost (Initial Year)	Description
Development Tools	Low / Free	Use of open-source frameworks such as Python, OpenCV, TensorFlow, and NLP libraries
Cloud Hosting	$60 – $80	Basic cloud server for backend API and database
Learning Content Production	$20 - $50	Creating diagrams and visual learning materials
Database Services	$30 – $50	Hosting MongoDB database
System Maintenance	$50 - $80	System updates and bug fixes

6.5 Pricing Strategy

The pricing strategy focuses on affordability so that educational institutions and students can easily adopt the system.

Possible pricing options include:

1. Free Basic Plan: Limited chatbot interaction and basic learning support.

2. Student Plan: Approximately $3–$5 per month for full access to chatbot tutoring and adaptive learning materials.

3. Institutional Plan: Approximately $200–$500 per year for schools that want to provide access to multiple students.

This pricing model ensures that the platform remains accessible while still providing sustainable funding for system maintenance and improvements.

6.6 Competitive Advantage

The proposed system offers several advantages compared with traditional e-learning platforms.

First, the system is specifically designed for deaf and hard-of-hearing students**,** focusing on visual explanations and simplified learning content.

Second, the integration of emotion detection technology allows the system to understand student emotional states and adjust learning materials accordingly.

Third, the system combines emotion detection, adaptive learning, and chatbot tutoring in a single platform, which is not commonly available in traditional e-learning systems.

These features provide a more personalized and engaging learning experience for students.

6.7 Intellectual Property Considerations

The proposed system includes several components that can be protected as intellectual property.

The software architecture, chatbot algorithms, and system implementation can be protected through software copyright protection**.** Additionally, custom-developed learning materials, diagrams, and educational content can also be protected under copyright laws.

If the system demonstrates significant novelty, future research may explore applying for patent protection for the emotion-aware adaptive learning methodology used in the platform.

Protecting the intellectual property of the system will help ensure ownership of the developed technology and support potential commercialization opportunities in the future.

7. BUDGET AND JUSTIFICATION

The development of the proposed Emotion-Aware E-Learning System requires several resources including hardware devices, software tools, cloud infrastructure, and human development effort. Although the project is developed within a university environment, it is important to estimate the actual cost of system development to understand the resources required if the system is implemented in a real-world environment. The following sections present the estimated budget and justification for each cost category.

Table 4: Budget

Category	**Item**	**Estimated Cost (LKR)**
Hardware	Laptop / Development Computer	250,000
Hardware	Webcam	8,000
Hardware	Internet Connectivity	6,000
Software	Open-source Development Tools (Python, OpenCV, TensorFlow, NLP Libraries)	0
Cloud Services	Cloud Hosting and Database Services	15,000
Human Effort	AI Model Development and Chatbot Implementation	200,000
Human Effort	Web Application Development and System Integration	120,000
Human Effort	System Testing and Evaluation	60,000
Total Estimated Budget		**659,000**

The budget mainly includes hardware resources, cloud services, and human development effort required to implement the system. A laptop is required for AI model development, software programming, and system testing. A webcam is necessary for capturing facial expressions used in the emotion detection subsystem. Cloud services are required to host the backend server, database, and learning materials so that the system can be accessed online. Human effort represents the

estimated cost of software development, including AI model implementation, chatbot development, web application development, and system testing. Open-source technologies such as Python, TensorFlow, OpenCV, and NLP libraries are used to reduce software licensing costs while still enabling advanced AI-based functionality.

8. WORK BREAKDOWN STRUCTURE (WBS)

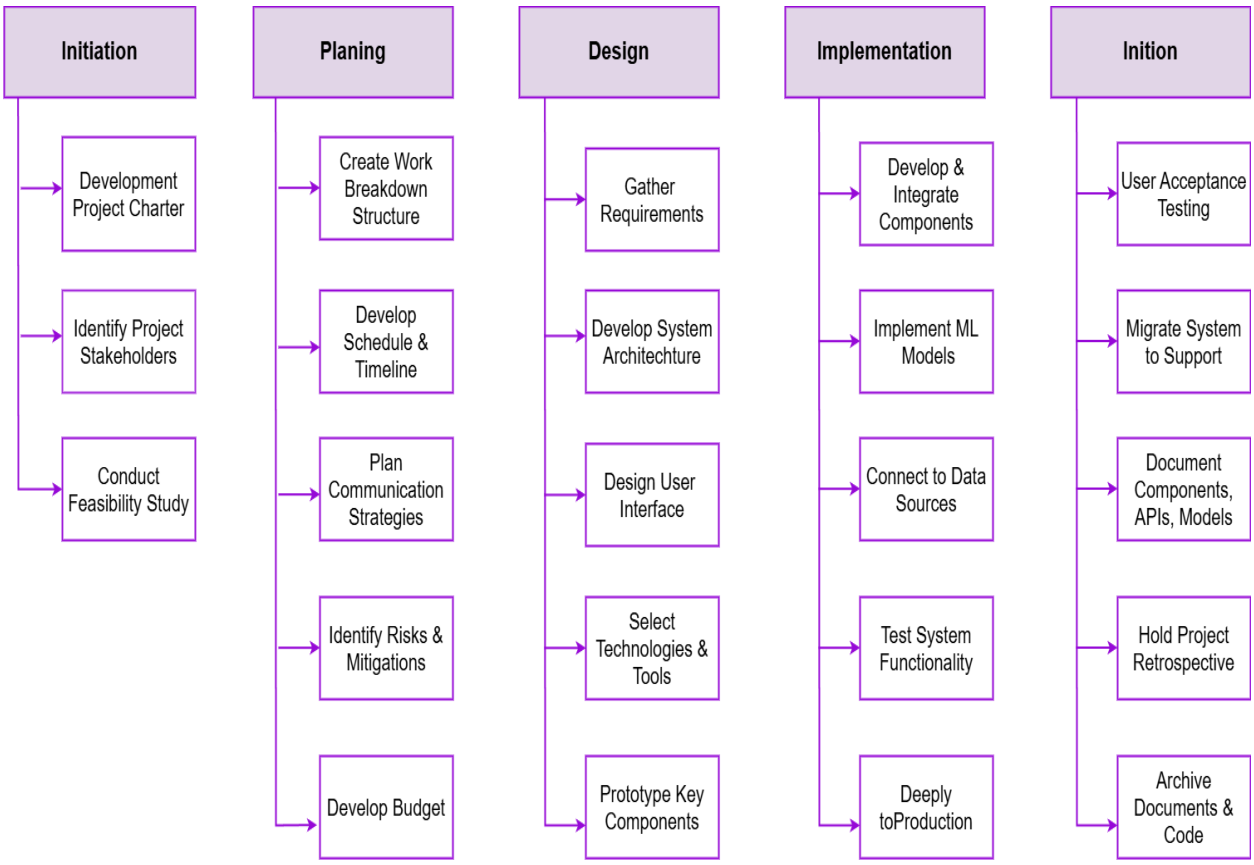


Figure . 2: Work Breakdown Structure

9. GANTT CHART

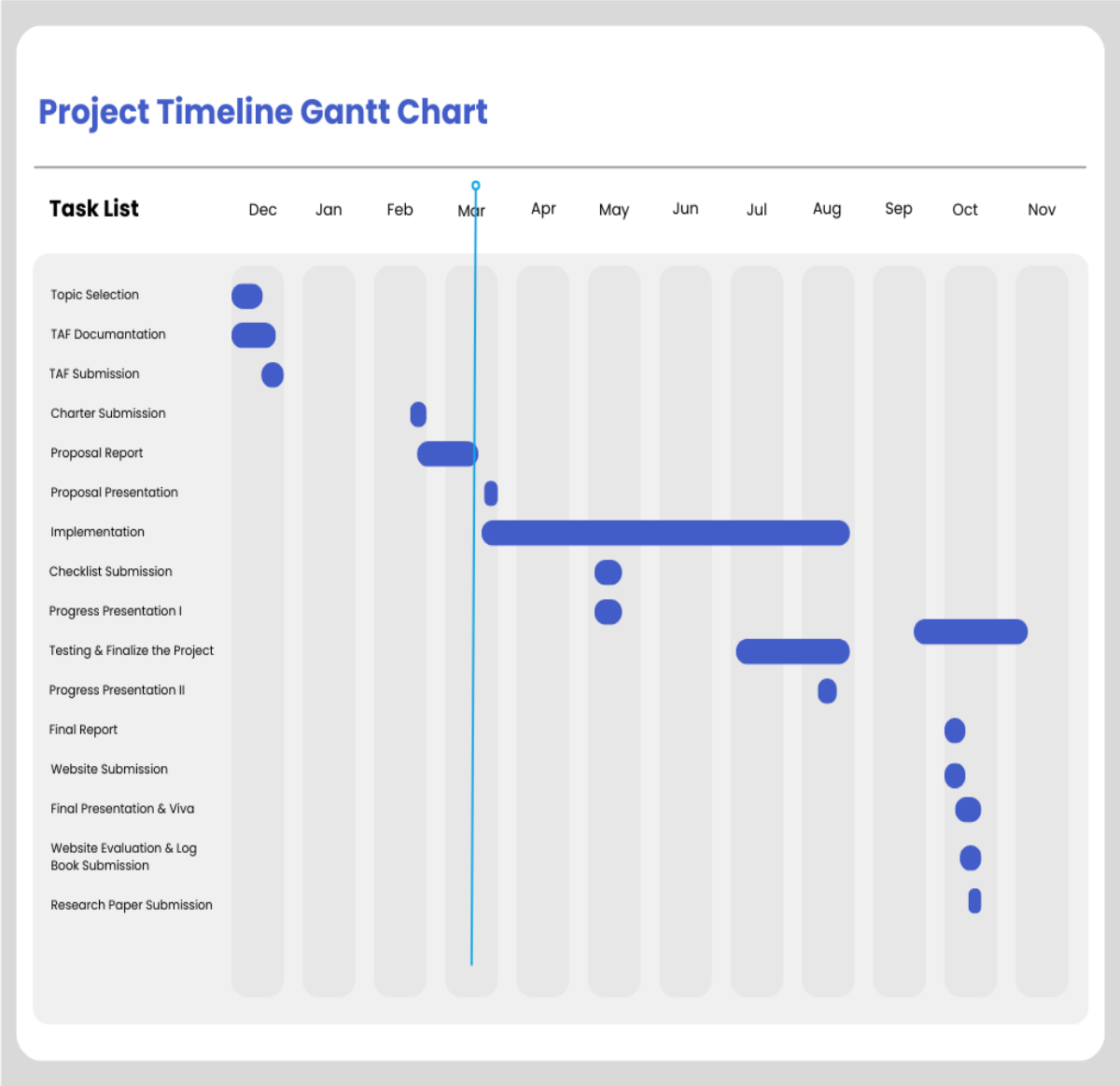


Figure . 3: Gantt chart

10. References

[1] R. Julianto, T. Gunawan, and E. A. Apriadi, “Development of an Intelligent Educational Chatbot Using NLP and Machine Learning,” *Int. J. Technology and Computer Science*, vol. 1, no. 1, pp. 14–24, Aug. 2025.

[2] A. Puspitasari, R. Kurniawan, and D. H. Santoso, “Natural Language Processing (NLP) Technology for Chatbot Website,” *Jurnal Penelitian Pendidikan IPA*, vol. 10, Special Issue, pp. 319–324, Aug. 2024. doi: 10.29303/jppipa.v10iSpecialIssue.8241.

[3] W. H. K. Fernando, P. T. E. W. Gunasekara, P. L. D. N. Perera, and H. Fernando, “Empowering Deaf Children with Sinhala Sign Language, Emotion Detection, and Sound Recognition,” in *Proc. 5th Int. Conf. Advancements in Computing (ICAC)*, 2023, pp. 1–6. doi: 10.1109/ICAC60630.2023.10417483.

[4] A. Mohamed, A. Elnekiti, T. M. Ghazal, K. Shaalan, and O. Alhyari, “Emotion-Aware Course Personalization in Virtual Classrooms via Multi-Modal Fusion Learning Systems,” in *Proc. 3rd Int. Conf. Cyber Resilience (ICCR)*, 2025, pp. 1–5. doi: 10.1109/ICCR67387.2025.11291875.

[5] M. M. Bala, H. Akkineni, and C. Srinivasulu, “An Approach for Learner Categorization Based on Emotions in Intelligent Adaptive E-Learning Environment,” *J. Web Eng.*, vol. 21, no. 5, pp. 1715–1734, July 2022.

[6] P. Kaushik, R. K. Singh, and S. Sharma, “Enhancing Emotional Support Chatbots Using Deep Learning & NLP: A Study on Personalization and Effectiveness,” in *Proc. 1st Int. Conf. Advances in Computing, Communication & Networking (ICAC2N)*, 2024, pp. 1–6. doi: 10.1109/ICAC2N63387.2024.10894854.

[7] Hand Talk, “Sign Language Translation App,” 2022. [Online]. Available: https://www.handtalk.me. [Accessed: Mar. 09, 2026].

[8] VLibras, “Brazilian Sign Language Mobile Application,” 2022. [Online]. Available: https://www.vlibras.gov.br. [Accessed: Mar. 09, 2026].

[9] Moodle, “E-learning Platform Accessibility Features,” 2023. [Online]. Available: https://www.moodle.org. [Accessed: Mar. 09, 2026].

[10] Microsoft, “Reading Progress Tool for Literacy Support,” 2022. [Online]. Available: https://www.microsoft.com/education. [Accessed: Mar. 09, 2026].

[11] L. Mahalakshmi, Y. A. Satyavati, E. Anbalagan, and V. H. Raj, “Emotion-Aware Sentiment Analysis in User-Generated Content via Attention-Augmented Deep Learning Frameworks,” in *Proc. 6th Global Conf. Advancement in Technology (GCAT)*, 2025, pp. 1–6. doi: 10.1109/GCAT66372.2025.11368444.